Write a Python Program to Implement Decision Tree

Aim

To develop a Python program that implements the Decision Tree algorithm for classification.

Objective

- To understand Decision Tree classification.
- To classify data using supervised learning.
- To train and test a Decision Tree model.
- To predict the output class for new data.

Theory

A **Decision Tree** is a supervised machine learning algorithm used for classification and regression problems. It consists of a root node, decision nodes, and leaf nodes. The algorithm splits the dataset into smaller subsets based on the best attribute selected using measures like **Information Gain** or **Gini Index**. Decision Trees are simple to understand, easy to visualize, and widely used in Artificial Intelligence and Data Mining.

Algorithm

Step 1: Import the required libraries.

Step 2: Prepare the training dataset.

Step 3: Create the Decision Tree classifier.

Step 4: Train the model using the training data.

Step 5: Predict the class for the test data.

Step 6: Display the predicted output.

Flowchart

Start

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Import Libraries

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Prepare Dataset

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Create Decision Tree Model

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Train the Model

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Predict Test Output

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Display Prediction

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Stop

Python Program

```
from sklearn.tree import DecisionTreeClassifier


X = [[0,0],
     [1,1],
     [1,0],
     [0,1]]


Y = [0,1,1,0]


model = DecisionTreeClassifier()
```

model.fit(X,Y)

prediction = model.predict([[1,1]])

print("Predicted Class =", prediction[0])

Sample Output

Predicted Class = 1

Result

The Python program successfully trained the Decision Tree model and predicted the output class.

Conclusion

Decision Trees provide a simple and efficient method for solving classification problems by learning decision rules from the training data.

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```
from sklearn.tree import DecisionTreeClassifier

# Training data
X = [
    [0, 0],
    [1, 1],
    [1, 0],
    [0, 1]
]

# Target labels
Y = [0, 1, 1, 0]

# Create the Decision Tree model
model = DecisionTreeClassifier()

# Train the model
model.fit(X, Y)

# Test data
test_data = [[1, 1]]

# Predict the class
prediction = model.predict(test_data)

# Display result
print("Predicted Class =", prediction[0])
```